

Tianrui (Dennis) Wu

Curriculum Vitae

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EDUCATION

JAN. 2026–MAY 2027

University of Arizona

Tucson, Arizona, USA

Bachelor of Science in Physics and Mathematics; expected May 2027.
GPA: 3.87/4.00. Advisor: Prof. Tim Eifler, Arizona Cosmology Lab.
Dean's List; Global Wildcat Scholarship (\$10,000/year merit award).

AUG. 2021–DEC. 2024

Duke University

Durham, North Carolina, USA

Coursework toward a Bachelor of Science double major in Physics and Mathematics with a minor in Computer Science; transferred, degree not conferred.
GPA: 3.92/4.00; Physics GPA: 4.00/4.00.
Advisors: Prof. Michael A. Troxel and Prof. Daniel Scolnic. Dean's List with Distinction.

PUBLICATIONS AND MANUSCRIPTS IN PREPARATION

Manuscript in Preparation

[1] **Wu, T.**, Paillas, E., et al. "BAO Reconstruction with Photometric Redshift Smearing." Manuscript in preparation (2026).

Peer-Reviewed Publications

- [2] OpenUniverse, LSST Dark Energy Science Collaboration, Roman Project Infrastructure Teams, Alarcón, A., et al. (including **T. Wu**). "OpenUniverse2024: A Shared, Simulated View of the Sky for the Next Generation of Cosmological Surveys." *Monthly Notices of the Royal Astronomical Society* **544**(4), 3799–3823 (December 2025). doi:10.1093/mnras/staf1833.
- [3] Riess, A. G., Scolnic, D., et al. (including **T. Wu**). "JWST Validates HST Distance Measurements: Selection of Supernova Subsample Explains Differences in JWST Estimates of Local H_0 ." *The Astrophysical Journal* **977**, 120 (2024). arXiv:2408.11770.

RESEARCH EXPERIENCE AND UNPUBLISHED WORK

JAN. 2026–PRESENT

BAO Reconstruction with Photometric Redshift Smearing

University of Arizona

Research Assistant, Arizona Cosmology Lab; mentor: Prof. Tim Eifler

- Lead a first-author study of baryon acoustic oscillation (BAO) reconstruction under photometric-redshift smearing as the sole analyst, coordinating with the principal investigator and a postdoctoral collaborator across the full pipeline from simulated-catalog generation through Bayesian posterior inference.
- Build a forward-model framework that propagates photometric-redshift uncertainty through reconstruction to quantify its effect on standard-ruler calibration for next-generation surveys including Roman and Rubin.
- Diagnosed and corrected a systematic bias in the Bayesian MCMC pipeline by cross-validating two covariance-estimation methods across a 250-cell sensitivity grid (5 noise levels $\times$ 5 data subsets $\times$ 2 model configurations), reducing the bias by sixfold.
- Mapped, using a 25-realization Monte Carlo ensemble, the smearing amplitude at which reconstruction's 20–40% precision gain becomes statistically insignificant; first-author manuscript in preparation.

JUN. 2024–AUG. 2024

Predicting Power-Spectrum Window Functions with Neural Networks

Caltech

Summer Undergraduate Research Fellow, SPHEREx Team; mentor: Dr. Olivier Doré

- Built a ResUNet convolutional neural network in PyTorch to deconvolve instrument window-function effects from galaxy power spectra, reaching sub-percentage prediction error on validation data.
- Owned the full machine-learning pipeline: lognormal mock-catalog generation with realistic SPHEREx survey masks, distributed training on Linux HPC, and validation against analytic baselines.
- Improved the analysis pathway for local primordial non-Gaussianity f_{NL}^{loc} and delivered a project proposal, two interim reports, a final presentation, and a final report to the SPHEREx faculty committee.

MAY 2022–DEC. 2024

Weak-Lensing Cosmology with Survey Synergies

Duke University

Research Assistant, Cosmology Group; mentor: Prof. Michael A. Troxel

- Engineered a Python pipeline for joint statistical analysis of 120 TB of Roman Space Telescope and Rubin Observatory simulations, constructing two-point correlation functions and using MCMC-based Bayesian inference to constrain cosmological parameters.
- Evaluated mode-based strategies and ground–space survey synergies for photometric-redshift calibration; the new methods increased the Figure of Merit by 20% and tightened constraints on cosmic homogeneity.
- Contributed simulation and analysis work to the OpenUniverse2024 collaboration and its MNRAS publication.

SEP. 2023–AUG. 2024

Distance-Ladder Calibration with Synthetic Stellar Populations

Duke University

Research Assistant, Cosmology Group; mentor: Prof. Daniel Scolnic

- Developed a Python pipeline for synthetic stellar catalogs and systematic feature extraction from Hertzsprung–Russell diagrams. Integrated novel distance-calibration methods into a unified framework for cross-team comparisons.
- Compared JWST distance measurements across the literature and quantified galaxy–dust correlations with current SDSS data using SQL and Python; contributed to the ApJ publication listed above.

MAY 2023–AUG. 2023

Quantifying Wetland Carbon Emissions in the Southeastern United States

Duke University

Data+ Team Lead; mentor: Prof. Wenhong Li

- Led a four-person team that applied climatology-based adaptive missing-value imputation and optimized a random-forest model by grid search; mentored teammates developing SVM and multilayer-perceptron models.
- Delivered a layered R Shiny analytics application and final research presentation.

INDUSTRY AND QUANTITATIVE RESEARCH

AUG. 2025–PRESENT

Quantitative Researcher and Systematic Trader

Origin Capital, Remote

Full-time Aug. 2025–May 2026; part-time May 2026–Present. Proprietary trading firm (\$35M).

- **Alpha and directional strategy:** Built an LSTM forecasting engine using phase-space features with a full MLflow MLOps lifecycle; integrated 200+ multi-frequency OHLCV, order-book, and derivatives factors with decay analysis and cross-sectional ranking, lifting live-model win rates by 3.1%. Deployed an HFT taker strategy on Coinbase International at a post-fee Sharpe of 8.2 over three months.
- **Execution and market making:** Architected TWAP, VWAP, and implementation-shortfall algorithms with microstructure timing overlays, reducing realized slippage by approximately 2 bps versus TWAP. Built vectorized and L2 queue-position backtesting engines used to validate a grid market-making strategy at a three-month out-of-sample Sharpe of 9.
- **Trading operations and risk:** Ran CTA books across OKX, Binance, and Coinbase while owning P&L monitoring, transaction-cost analysis, and fee analysis. Directed multi-strategy risk across \$35M in prop/client capital using Grafana monitoring, leverage, position, and beta limits plus pre/post volatility and model attribution; trimmed exposure ahead of the October market crash.
- **Research leadership:** Led a two-person research pod across production code, factor research, microstructure analysis, market-making models, and strategy development; mentored junior researchers and served as the primary methodology and code reviewer before live deployment.

Applied technologies: Python (PyTorch, MLflow, Polars, pandas, NumPy), CEX REST/WebSocket APIs, Linux, Git.

FEB. 2025–AUG. 2025

Quantitative Research Intern

InsightCheck Investments, Shenzhen, China

Systematic equity fund with CTA desk (\$300M assets under management).

- Developed a market-neutral long/short equity strategy combining PCA-derived statistical factors, mean-reversion signals, and a covariance-based risk model, backtested at a post-cost Sharpe of 2.4.
- Researched fundamental and microstructure alpha factors for Chinese A-shares, validating 62+ intraday factors with daily information coefficient above 0.01 and conducting cross-sectional portfolio optimization in Polars.
- Solely engineered an L3 market-by-order reconstruction engine, prototyping in Python and migrating to C++, achieving approximately one second per trading day for the full A-share universe.
- Built a Dask-based distributed backtester that shortened evaluation cycles tenfold to approximately 15 minutes.

Applied technologies: Python (Polars, Dask), C++, SQL, Linux, Git.

SELECTED QUANTITATIVE PROJECTS

2026	IMC Prosperity 4 Trading Competition Ranked in the top 3% of 6,000+ teams worldwide by developing an automated trading bot using behavioral fingerprinting of counterparties, fair-value estimation, market making, and cross-asset arbitrage.	<i>Python (PyTorch), Rust</i>
2025	Crypto Market Maker and Backtest Engine - Built and deployed a live ETH market-making framework on OKX that combined EWMA flow alphas with a CNN capturing order-book patterns through phase-space features, supported by a WebSocket data feed and Grafana monitoring. - Averaged 0.5–1% daily return; the project led to recruitment at Qrigin Capital.	<i>Python, Rust</i>

AWARDS AND FELLOWSHIPS

2026	National First Place, U.S. Physicists' Tournament.	
2026	Global Wildcat Scholarship, University of Arizona. \$10,000/year merit scholarship for international transfer students.	
2024	Faculty Scholars Finalist, Duke University. Sole physics nominee; among Duke's highest undergraduate honors.	
2024	Jocelyn Bell Burnell Outstanding Leadership Scholarship, Society of Physics Students. \$6,000; sole recipient nationwide.	
2024	Inductee, Sigma Pi Sigma National Physics Honor Society. Membership awarded for academic distinction in physics.	
2023	Outstanding Chapter Award, Society of Physics Students. Awarded to the Duke chapter during Wu's presidency.	
2021	Third Grand Award, Physics and Astronomy, Regeneron International Science and Engineering Fair. \$1,000 award in the Physics and Astronomy category.	
2021	Top Gold Medalist, British Physics Olympiad. Ranked in the national top ten.	
2021–2026	Dean's List with Distinction / Full-Time Dean's List. Duke University (Fall 2021, Spring 2023, Fall 2023, Fall 2024) and University of Arizona (Spring 2026).	

TALKS AND PRESENTATIONS

JUN. 2026	BAO Measurements, Cosmology Session. <i>SPHEREx Science Team Meeting (Face-to-Face E4), IPAC, Caltech.</i>	
APR. 2026	BAO analysis segment of the SPHEREx team colloquium led by Prof. Elisabeth Krause and Prof. Tim Eifler. <i>SO/NSF NOIRLab Joint Colloquium, Steward Observatory, University of Arizona.</i>	
AUG. 2024	"Predicting Power Spectrum Window Function with Neural Networks." <i>SURF Final Presentation, SPHEREx Team, Caltech.</i>	

TEACHING

FEB. 2025–AUG. 2025	Physics and Mathematics Tutor Taught high-school physics and mathematics for 20 hours/week, including AP Physics, AP Calculus, AP Mathematics, and Cambridge A-Level Physics; guided more than 15 students to the top AP exam score of 5.	<i>PalmDrive Academy</i>
AUG. 2022–APR. 2024	Teaching Assistant and SAGE Facilitator Supported Multivariable Calculus, Linear Algebra, and Advanced Probability. Assisted 40+ students weekly across two three-hour Help Room shifts and led two one-hour SAGE discussion sections each week. Rated "Excellent" (30/30) by supervisors and peers in the 2023 and 2024 reviews.	<i>Duke University Mathematics Department</i>

LEADERSHIP AND OUTREACH

SEP. 2021–DEC. 2024	Founder and President, Stargazing Devils Astronomy Club <i>Duke University</i> - Founded Duke’s first astronomy club and organized weekly observatory open houses with 100+ attendees, three comet events, four meteor-shower events, and three eclipse events; served as a NASA Partner Eclipse Ambassador. - Secured funding for 14 students to attend the 2023 and 2024 solar eclipses, organized a campus eclipse watch party for 300+ participants, and led 20+ astrophotography events using the Skynet platform.
JUN. 2022–DEC. 2024	President (2024), Co-President (2023), Society of Physics Students <i>Duke University</i> - Revived the chapter after the pandemic through four research panels, six student–faculty seminars, three laboratory tours, MATLAB/LaTeX workshops, graduate-school panels, and a women’s mentorship program. - Organized the 2023 SPS Zone 5 meeting for 100+ participants; the chapter received the 2023 Outstanding Chapter Award.
JUN. 2022–MAY 2024	Student Ambassador, American Physical Society <i>Remote</i> - Recruited APS members; organized physics study sessions, manuscript workshops, and publication events; served as Technology Chair for the Physics Undergraduates Learning and Sharing Experiences webinar series. - Attended the 2023 and 2024 APS Leadership Meetings and advocated on physics-community priorities during U.S. Congressional Visits Day.
SEP. 2021–AUG. 2024	Senior Editor and Peer Reviewer, Duke Vertices Science Journal <i>Duke University</i> Reviewed two STEM manuscripts per month and implemented a synopsis system for general audiences that increased website traffic by 210%; invited by the editor-in-chief to join the executive team.

TECHNICAL SKILLS, COURSEWORK, LANGUAGES, AND INTERESTS

PROGRAMMING	Python (expert), Rust, C/C++, Java, R, SQL/PostgreSQL, MATLAB, Julia, JavaScript, and L ^A T _E X.
MACHINE LEARNING AND DATA	PyTorch, MLflow, Polars, pandas, NumPy, Numba, Dask, qlib, SciPy, astropy, Matplotlib; ResUNet/CNN and LSTM architectures; random forests, support-vector machines, multilayer perceptrons, feature engineering, selection, and grid-search optimization.
STATISTICAL AND SCIENTIFIC METHODS	Bayesian inference, MCMC, Monte Carlo simulation, statistical modeling, sensitivity analysis, covariance estimation, two-point correlation functions, information-coefficient validation, distributed training and backtesting, Linux HPC, Git, REST/WebSocket APIs, and Grafana monitoring.
DOMAIN KNOWLEDGE	Observational cosmology (BAO, weak lensing, power spectra, photometric-redshift calibration), market microstructure, live strategy development and backtesting, optimal execution, factor research, digital assets, commodity futures, global equities, options (Akuna 101/201 completed), and prediction markets.
SELECTED COURSEWORK	Stochastic Calculus; Statistical Machine Learning; Advanced Mathematical Modeling; Computational Physics; Real Analysis; Advanced Probability; Statistical Learning and Inference; Graduate Machine Learning; Probability; Data Structures and Algorithms; Microeconomics; Macroeconomics; Game Theory; Cosmology*; General Relativity*; Stellar Astrophysics*; Quantum Theory; Electrodynamics; Abstract Algebra*; Differential Geometry. *Graduate-level course.
LANGUAGES	English (fluent); Mandarin Chinese (native).
INTERESTS	Personal trading portfolio (AI-assisted, factor-driven strategy with discretionary macro and catalyst overlays); poker; competitive GeoGuessr; tabletop wargaming; strategy board games; chaos theory; forecasting; astrophotography; cosmology; and science fiction.